Gesture-Based Assistive Communication Using TinyML and FSK for Real-Time Embedded Systems

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***Abstract*—In this paper, a portable, low-cost assistive communication system is presented, which is suitable for people with speech or motor disabilities. In this system, a MPU6050 sensor has been used to identify hand gestures, which are then classified in real-time using TinyML models. The detected gesture information is sent wirelessly using Frequency Shift Keying modulation, which is then received at the receiver end, decoded, and presented through a Python-based interface. MATLAB has been used to analyze the received signals, creating a better understanding of the waveforms, thereby increasing the accuracy of detection. This method provides offline, real-time communication, which is beneficial in enhancing the lives of people with disabilities.**

I. Introduction

The rapid advancement in wearable technology and embedded sensor systems has led to new opportunities for improving human-computer interaction. Among these emerging technologies, gesture-based interaction systems have gained significant popularity due to their ability to offer users a more natural way of interacting with computing systems. A prime example of this trend is the Gesture-Based Assistive Communication System, which utilizes motion sensors, TinyML, and FSK communication protocols for recognizing hand gestures and converting them into meaningful information in the form of text or speech.

The main component of this communication system is a motion sensor known as MPU6050, which consists of both an accelerometer and a gyroscope for accurately recognizing hand gestures. The sensor is embedded in a wearable device for recognizing hand gestures performed by users. The ESP32 microcontroller is used for processing sensor information, utilizing TinyML for recognizing gestures, and sending information through a Frequency Shift Keying (FSK) protocol for communication with a receiver. At the receiving end, a Python-based interface is used for visualizing gestures and converting them into speech for communication by users who suffer from speech or motor disorders.

The importance of the system is based on the fact that it is capable of providing an affordable, portable, and real-time assistive communication system that can be used entirely offline without the need for cloud-based computation. This project is based on the integration of sensor technology, machine learning, and digital communication, providing a platform that can be used to improve the life of people with disabilities.

This study is based on the exploration of the technological aspects of the system, including the sensor integration, sensor processing, gesture recognition using TinyML, and FSK-based communication. In addition, the study is based on the exploration of the potential applications of the system in the real world, including the importance of wearable devices in the real world and the significance of innovation in wearable technology.

II. Literature Review

Over the past few years, there has been substantial research in the development of wearable technology for gesture recognition, especially for assistive communication purposes. In the initial stages of the study, there was the usage of gyroscopes and accelerometers for the detection of hand movements. This was followed by the analysis of the data collected from the sensors to recognize the gestures. Some of the examples of the usage of gesture recognition include the development of data gloves and the Wii Remote. However, there have been challenges in the accuracy of the results obtained from the sensors.

The development of sensor technology has been one of the most important factors in the development of the precision of gesture recognition systems. In recent times, the MPU6050 sensor has been widely used for the purpose of gesture recognition. It is a combination of the accelerometer and the gyroscope sensor. It has been observed that the usage of sensor fusion techniques and the development of proper programming algorithms have resulted in the precision of the results obtained from the sensors.

At the same time, the developments in machine learning, especially lightweight machine learning, have enabled the shift from the offline post-processing of gestures to real-time computation. TinyML has enabled the ESP32 microcontroller to perform complex computations on the embedded system itself, eliminating the constant need for connectivity to the cloud. Various recent studies have proven the effectiveness of the integration of TinyML with wearable devices not only in terms of real-time feedback but also in terms of the cost-effectiveness of the assistive system.

Recent studies have highlighted the significance of real-time feedback in the development of assistive systems based on gestures. Mobile applications have been designed to provide real-time feedback to the user in the form of visual and auditory feedback. Various studies have proven the effectiveness of the combination of wearable devices with interactive software in the development of assistive systems for people

with disabilities. It has been proven that gestures can be translated into text or speech through the combination of wearable devices with interactive software.

Overall, these developments highlight the promise of sensor technology, machine learning, and interactive technology in delivering advanced assistive technology solutions. Leveraging previous research in motion sensing, embedded technology, and feedback, gesture recognition technology can now deliver more accurate, user-friendly, and pragmatic solutions for facilitating communication for those with impairments.

III. Methodology

A. System Architecture

The system architecture of the proposed system for gesture-to-speech communication includes the MPU6050 sensor for detecting the gestures by considering the acceleration and angular velocity of the moving object using the accelerometer and gyroscope sensors. The ESP32 microcontroller is programmed using the Arduino IDE and reads the data from the sensor. It performs the initial processing of the data by filtering the noise from the sensor data. Then the data is sent to the computer system.

The MATLAB processing system reads the data from the microcontroller and uses the energy-based gesture detection technique to analyze the gestures and detect the valid hand movements. Once the valid gestures are detected by the system, they are converted into a predefined text message representing the communication. The MATLAB system uses the Windows Speech Synthesizer interface to convert the text into speech.

In addition to the speech generation from the detected message, the detected message is converted into a binary format and modulated using the Frequency Shift Keying (FSK) technique for the generation of the communication signal. The MATLAB system uses the time domain plot and the spectrogram technique to visualize the detected gestures and the modulated communication signal.

B. Block Diagram

The block diagram of the proposed gesture-to-speech communication system illustrates an organized framework that incorporates motion sensing hardware, embedded processing, and MATLAB-based signal analysis, enabling hand gestures to be translated into sound and communication signals. This proposed gesture-to-speech communication system incorporates an MPU6050 motion sensing hardware, an ESP32 microcontroller, and MATLAB-based processing.

At the sensing level, an MPU6050 motion sensing hardware is utilized, and it captures hand gesture information. This MPU6050 motion sensing hardware incorporates an accelerometer and a gyroscope, and these two sensors can effectively detect linear and angular movements in the hand. This information provides detailed information about hand gestures, including orientation and dynamic motion characteristics. This sensor continuously measures raw data that corresponds to hand gestures made by users.

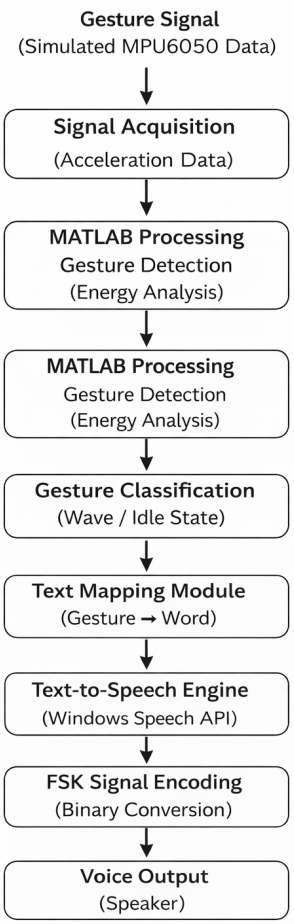

Fig. 1. System architecture of gesture-to-speech communication system

The ESP32 microcontroller is an embedded processing hardware that functions as the core processing element in the proposed gesture-to-speech communication system. It communicates directly with the MPU6050 motion sensing hardware via an I2C communication protocol and captures motion data in real time. This microcontroller is programmed using an Arduino IDE.Within the firmware, the raw sensor readings are filtered to eliminate any noise and stabilize the signal measurements. Basic preprocessing and calibration are performed on the raw sensor readings to ensure accurate representation of the motion data.

Once the preprocessing is complete, the ESP32 sends the captured gesture data to a computer system. The ESP32 uses a serial communication method to send the sensor readings to the MATLAB system. Within the MATLAB system, the captured gesture signals are processed. The waveform of the motion is analyzed to observe the gestures. An energy detection method is employed to identify the occurrence of a valid gesture. Within the MATLAB system, the energy of the signal is calculated and compared with a predefined threshold value to differentiate gestures from idle states.

Upon the occurrence of a valid gesture, the system classifies the gestures by mapping the identified motion pattern to a predefined text message. Every gesture is mapped to a predefined word or command. This process converts the gestures into a digital text format that can be utilized for communication. The recognized text message is then processed in a speech synthesizer module. MATLAB interacts with the Windows Speech Synthesizer using the .NET framework and incorporates a text-to-speech (TTS) engine. The text message is converted into speech and played through the system speaker.

In addition to speech output, the system also demonstrates the generation of a communication signal using the text message in binary form using ASCII code. The binary message is modulated using the frequency shift keying (FSK) technique. In the FSK technique, two different carrier frequencies represent binary '1' and binary '0.' MATLAB generates the FSK waveform corresponding to the binary message and displays the modulated signal in the time domain.

The generated signal can be further analyzed using a spectrogram visualization tool. The tool shows the frequency transitions during the modulation process. It can be noted that the frequency transitions during the modulation process are valid, proving the implementation of the FSK technique.

The above description illustrates a block diagram of a system that incorporates motion sensing hardware, embedded systems using Arduino-based firmware, and MATLAB-based signal processing. It demonstrates the feasibility of a gesture-based speech communication system and the potential of using embedded systems with high-end signal processing platforms.

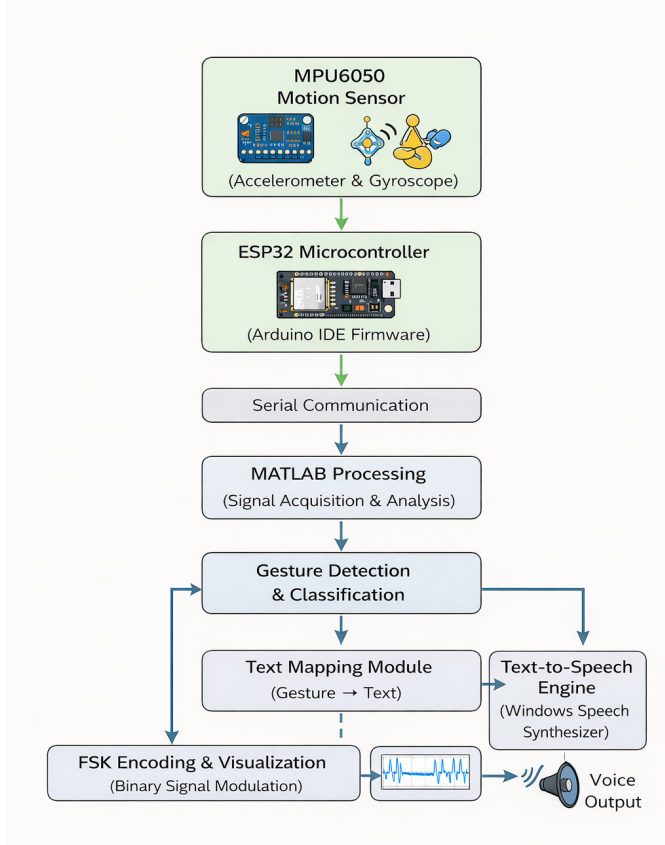

Fig. 2. Block Diagram Of Gesture-To-Speech Communication

The methodology adopted for developing the gesture-to-speech communication system is as follows:

Firstly, during the signal generation phase, a simulated waveform is generated for the gesture motion within a MATLAB environment. Since the system is intended to mimic a signal generated by accelerometers for motion detection, similar to those generated by MPU6050 devices, a sinusoidal waveform is adopted along with some level of random noise to mimic hand movement patterns. The above-mentioned simulated waveform is intended to mimic a realistic signal that could be generated by motion sensors like MPU6050 devices.

Secondly, during the signal processing and gesture detection phase, the system is intended to analyze the motion signal generated earlier to ascertain whether a gesture has been performed or not. The MATLAB environment calculates the total amount of energy within the signal by summing up squared values within the signal. If the amount of energy is found to be greater than a predefined threshold value, it is ascertained that a gesture has been performed; otherwise, it is ascertained that an idle signal is being generated.

Thirdly, the classification of the gestures and the text mapping are done once a valid gesture is detected. The detected gesture is mapped to a predefined word or message stored in the system. This process converts the detected gestures into text that can be processed for communication purposes.

Fourthly, the speech synthesis is done by integrating the Windows Speech Synthesizer tool with MATLAB. The text mapped from the detected gestures is sent to the text-to-speech tool. The text is converted into voice using the text-to-speech tool. This enables the system to generate voice corresponding to the detected gestures. This helps the system to communicate through gestures.

Lastly, the system uses the concept of digital communication by converting the text generated from the detected gestures into binary format. The text is modulated using Frequency Shift Keying (FSK) to generate a modulated waveform. In FSK modulation, the binary values are represented by two different frequencies. MATLAB is used to generate the modulated waveform. The waveform is analyzed by plotting the time-domain plots and the spectrogram.

Testing and visualization are conducted to ensure proper operation of each stage. MATLAB plots are employed to analyze gesture signals, test the performance of the proposed energy detection method, and check the frequency transition of generated FSK signals. By passing through these stages, it is evident that the proposed system is able to convert gesture signals into speech as well as digital communication signals.

IV. Results And Discussion

A. Hardware and Software Integration

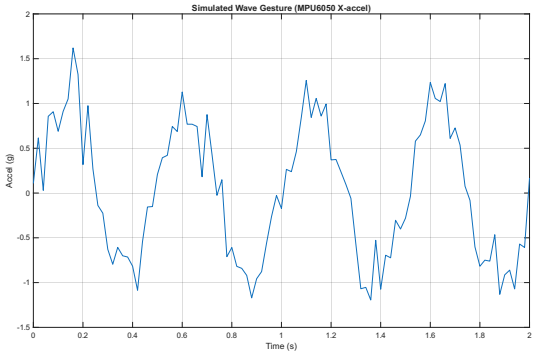

Fig. 3. Simulated Gesture Signal From MPU6050 Accelerometer

In the proposed system, there is integration of gesture sensing, signal processing, and speech generation using the programming tool MATLAB. In the proposed system, gesture data is simulated using a waveform similar to that obtained from an MPU6050 sensor's accelerometer. The waveform is a sinusoidal waveform with added white noise, simulating real-world motion patterns. In the proposed system, the signal is processed using MATLAB to check for the energy content of the signal and determine whether there is a gesture. Once the gesture is identified, it is converted into speech using the Windows Speech API. In addition, the identified message is converted into a FSK signal for analysis.

B. Gesture Recognition and System Performance

In gesture recognition, a signal energy threshold method is used to identify active gestures from idle states. Once the energy level is found to be above a predefined threshold level, it is considered a valid gesture. From the experimental results, it is evident that the simulated waveform effectively represents gesture activity and is easily recognized using this method. It is also evident that the FSK signal is effectively switching between the binary data frequencies, thus proving that the FSK modulation has been correctly implemented.

C. Voice Output and System Interaction

Once the gesture is detected, it generates speech output for the recognized gesture through MATLAB's integration with the Windows Speech Synthesizer. The gesture is converted into a word and spoken through the system's speaker in real time. This shows how gesture recognition can be used for voice output and how it is feasible to create assistive communication systems for people with disabilities through gesture recognition.

V. Conclusion

In order to recognize and interpret hand gestures and provide real-time voice feedback for the process of communication, the proposed gesture-based assistive communication system has effectively integrated signal processing and speech synthesis using MATLAB. Gesture signals, similar to those used by motion sensor devices, are processed using energy-based detection mechanisms for recognizing significant hand movements. After recognizing the gesture, it is converted into defined text and speech using the MATLAB.

In addition, the proposed system has also been used to demonstrate digital communication concepts by generating FSK signals for the detected message, thereby visualizing the process of signal transmission. The results obtained through the proposed system demonstrate that the gesture activity is detected and converted into voice output with minimum delay.

Overall, the proposed gesture-based assistive communication system has effectively demonstrated the process of gesture recognition and conversion into speech, thereby providing a potential solution for assisting individuals with speech impairments through more natural and accessible interaction with technology.

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